

LLM Baseline Evaluation

GNN Coding Benchmark (GNN-CB)

May 2026

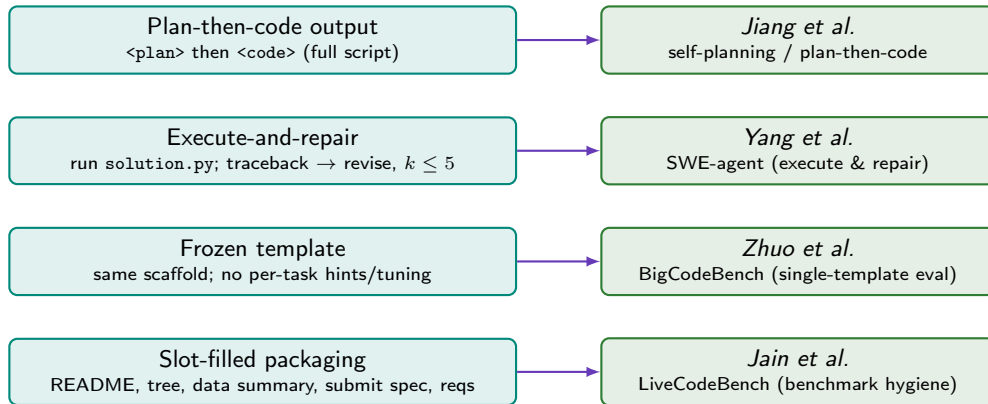
“The art of programming is the art of organizing complexity.”

— Edsger W. Dijkstra

Protocol Components and Reference Papers

Fixed protocol components (prompt.md)

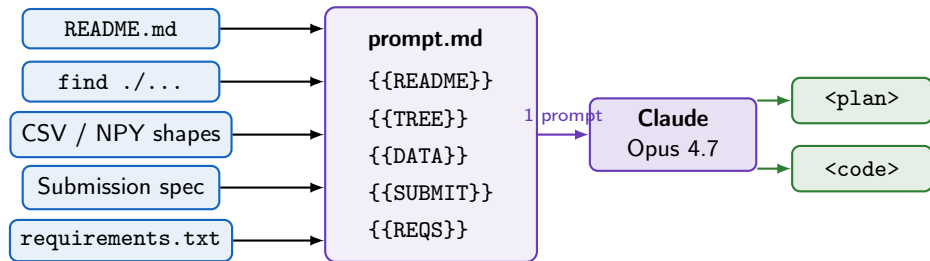
Where each idea comes from



Key claim (discovery paper)

We benchmark **humans vs LLMs** under a **controlled protocol**; differences reflect model behavior rather than prompt-engineering effort.

How `prompt.md` Instantiates Per Competition



Same template

`prompt.md` is unchanged across all competitions. Only slot *contents* differ.

No human judgment

All slots filled by mechanical rules (no curation, no hints).

Reproducible

Any researcher can reproduce results with only `prompt.md` + repos.

Structure of a Generated Plan

The LLM emits a **structured reasoning trace** *before* writing code — forcing architectural decisions upfront.

Plan-then-code pattern: Jiang et al. TOSEM 2024

- 1 **Task framing** — what to predict
- 2 **Feature engineering** — node / graph features
- 3 **Architecture** — backbone, pooling, head
- 4 **Training protocol** — loss, optimizer, reg.
- 5 **Validation & threshold** — metric, split
- 6 **Submission format** — path, column names

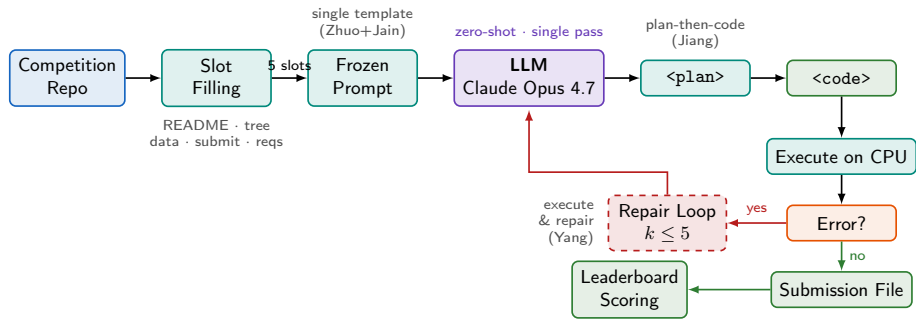
Why it matters

Reduces hallucinated APIs and wrong architectures by separating design from implementation.

Example — Real_Or_Fake Plan (C03, Easy)

1. Features: BERT(768) + spaCy(300)
+ profile(10) -> 1078-dim/node.
Baseline: spaCy only.
2. Profile: logit + StandardScaler
(fit on training nodes only).
3. Arch: 2-layer GraphSAGE hidden=128
-> mean+max pool -> MLP head.
4. Root-node pathway: separate MLP
concat before classifier head.
5. Training: Adam lr=1e-3, BCE,
batch=64, early stop (patience=5).
6. Threshold: sweep [0.30,0.70]
step 0.01 -> t=0.53 on val F1.
7. Submit: predictions.csv (id,y_pred)
Self-check: row count + ID match.

Evaluation Architecture



Design Principles

- Single frozen template for all 18 competitions
- No per-task hints, examples, or prompt edits
- Up to 5 repair iterations on runtime error
- Scoring via each repo's own evaluation script

Methodological Lineage

- <plan>+<code> Jiang et al. TOSEM 2024
- Execute-and-repair Yang et al. NeurIPS 2024
- (SWE-agent) + Shinn et al. 2023 (Reflexion)
- Single-template Zhuo et al. 2024 (BigCodeBench)
- + Jain et al. 2024 (LiveCodeBench)

System Prompt — identical for all 18 competitions

Given a GNN coding competition:

- (1) competition README,
- (2) repository file tree,
- (3) data-sample summary,
- (4) required submission format.

Produce a self-contained Python script that trains a GNN and writes the submission file.

Respond EXACTLY as:

```
<plan> 5-10 bullet plan </plan>
<code> complete script </code>
On error: return revised <plan>
and <code>. Max 5 repairs.
```

Plan+code: Jiang et al. TOSEM 2024; Repair loop: Yang et al. NeurIPS 2024 (SWE-agent), Shinn et al. 2023 (Reflexion)

User Prompt — 5 Mechanical Slots

Slot	Filled by
{{README}}	Verbatim README.md
{{TREE}}	find . -type f
{{DATA}}	shapes + .head(5)
{{SUBMIT}}	Submission spec
{{REQS}}	requirements.txt

Key Constraint

Template frozen on one pilot (C03) before remaining competitions. No human judgment in slot filling.

Zhuo et al. 2024 (BigCodeBench); Jain et al. 2024 (LiveCodeBench)

Task

Binary graph classification for **fake news detection**.

Dataset: GossipCop Metric: **Binary F1**

LLM Solution (from plan.md)

- **Features:** BERT (768) $\oplus$ spaCy (300) $\oplus$ profile (10) $\rightarrow$ 1078-dim (*baseline: spaCy only*)
- **Model:** 2-layer GraphSAGE, hidden=128
- **Pooling:** mean + max $\rightarrow$ MLP classifier
- **Root pathway:** separate MLP for root node
- **Threshold:** sweep [0.30, 0.70] on val $\rightarrow$ **t = 0.53**

Results

Validation Binary F1	0.9768
Validation Accuracy	0.9762
Wall-clock (CPU)	$\approx$ 2 min
Repair iterations	0

Leaderboard snapshot (test F1)

#	Team	F1
1	samuel	0.9803
–	Claude 4.7 (val)	0.9768
2	Murad	0.9782

Takeaway

Within **0.4 pp** of human #1 — zero repairs.

Task

Link prediction on CiteSeer citation network.

Metric: **ROC-AUC**

LLM Solution (from plan.md)

- **Graph:** positive edges for message-passing only
- **Features:** 2742-dim sparse BoW, L2 row-norm
- **Encoder:** 2-layer GraphSAGE, hidden=128
- **Decoder:** dot-product $\langle z_u, z_v \rangle \rightarrow \text{sigmoid}$ (Kipf & Welling 2016)
- **Training:** Adam, full-batch, 200 epochs, early stop AUC

Results

Validation ROC-AUC	0.6934
Avg. Precision	0.6850
Wall-clock (CPU)	23 s
Repair iterations	0

Leaderboard (test AUC)

#	Team	Model	AUC
1	VinitSingroha	GAT	0.8914
2	Gurur Gamgam	GAT	0.8898
...	...		...
–	Claude 4.7 (val)	SAGE	0.6934

Takeaway

Mid-pack. Correct SAGE recipe; top humans used GAT + structural features.

Task

7-class node classification on the Cora citation network.
 Features: 1433-dim BoW + **Gaussian noise** ($\sigma=0.4$) applied.
 140 train · 500 val · 1000 test nodes Metric: **Accuracy**
Model: GPT-5.5, zero-shot, CPU-only

LLM Solution (from plan.md)

- **Features:** standardized 1433-dim BoW; symmetric + RW normalization
- **Backbone:** SGC — linearized GCN via $A^K X$ propagation, $K \in \{2, 3, 4\}$
- **Classifiers:** one-vs-rest logistic regression, $C \in \{0.01, 0.03, 0.1\}$
- **Smoothing:** label propagation, $\alpha \in \{0.90, 0.95, 0.98, 0.99\}$
- **Decoding:** average 13 probability outputs $\rightarrow$ argmax over 7 classes

Results (validation set, 500 nodes)

Validation Accuracy	73.20%
Best single SGC	71.20%
Best label propagation	71.20%
Wall-clock (CPU)	0.87 s
Repair iterations	0

Leaderboard (test Accuracy)

#	Team	Acc
1	TugaAhmed	74.20%
–	GPT-5.5 (val)	73.20%
2	gururgg	70.90%
3	abdksm	70.70%

Takeaway

Within **1 pp** of human #1. Ensemble of SGC + label propagation

Task

3-class: *organic* / *grid* / *hybrid* street-network layouts.

Dataset: OpenStreetMap Metric: **Macro-F1**

84 train + 36 test graphs.

LLM Solution (from plan.md)

- **Node features:** x , y , street_count, degree, clustering (5-dim)
- **Graph stats:** density, degree_var + **orientation entropy** (Boeing 2019)
- **Backbone:** 2-layer GIN + mean/max pool $\oplus$ graph stats $\rightarrow$ MLP
- **Training:** weighted CE, 5-fold stratified CV
- **Inference:** soft-vote ensemble of 5 fold-best models

Results (5-fold CV, 84 training graphs)

OOF Macro-F1	0.7167
CV mean $\pm$ std	0.7184 ± 0.0672
Wall-clock (CPU)	9 s
Repair iterations	0

Leaderboard snapshot (test Macro-F1)

#	Team	Macro-F1
–	Claude 4.7 (OOF)	0.7167
1	ldrees_submission	0.5368
2	tasneem	0.5021
...		

Takeaway

Exceeds human #1 by +0.18. Orientation entropy absent from all human baselines.

Summary Results

#	Competition	Difficulty	LLM Score	Human Top	Gap
C03	Real_Or_Fake	Easy	0.9768 (<i>val F1</i>)	0.9803	−0.0035
C09	NetLinkArena	Med	0.6934 (<i>val AUC</i>)	0.8914	−0.1980
C02	CGCC	Hard	0.7167 (<i>OOF F1</i>)	0.5368	+0.1799
C07	GNN_CoRA	Med	0.7320 (<i>val acc</i>)	0.7420	−0.0100

Easy — Near Parity

Within 0.4 pp of human #1.

Well-templated tasks are solvable by frontier LLMs.

Medium — Mid-Pack

Correct recipe; missed GAT + structural features used by top humans.

Hard — Exceeds #1

Domain prior knowledge (orientation entropy) offsets small training pool.

Thank You